

Justifying cvc5 rewrite holes with egglog and RARE: run **rw2** on QF_UF, QF_LIA and QF_LRA, and the state of the tooling

Carcara branch `egglog/rewrite-holes`

Run of 2026-09-24/25 (binary `3f982c71`) – report of 2026-09-25 (tooling at `fedf0b31`)

Addendum §8: run **rw3** and the `ds11` comparison arm of 2026-09-25/26

Addendum §9: runs **rw4** and **rw5** of 2026-09-27/28

Abstract

cvc5's Alethe proofs leave the rewrites its post-processor cannot expand as premise-free hole steps tagged `MACRO_REWRITE` or `MACRO_SR_PRED_INTRO`. Carcara's hole pipeline hoists and prunes the repeated ones, tries each with a checker-certified normalizer and, where that does not close it, with egglog saturation over a RARE rule set, reconstructs an Alethe subproof from the saturated e-graph, and re-checks the result. This report summarises one run over 7,381 benchmarks of QF_UF, QF_LIA and QF_LRA, every one with a complete cvc5 proof (4,500,561 hole steps, 2,319,574 distinct holes after hoist and prune). Of the holes attempted, 99.5% are justified with Alethe steps and 99.8% are proved (justified, or proved by egglog and lost in the reconstruction); the normalizer alone settles 30.0% of them before egglog runs. 4,496 proofs come out with every hole justified and 1,506 re-check `valid`; the gap between the two is the 3,358 proofs that carry a hole the pipeline never attempts (cvc5's untagged arithmetic and subtype-elimination trust steps), plus 316 proofs in which a certificate itself falls back to a trust step. The per-proof budget, which dominated the previous run, binds 543 holes. The 11,980 kept holes are: 35% egglog time or memory kills, two thirds of them in four QF_LRA families; a third the reconstruction's own losses on QF_UF's coarse Boolean holes (no certificate, or the search killed after egglog had proved the goal); and a small honest residue of goals outside the rule set. The reconstruction and elaboration causes were reproduced after the run and are fixed at `fedf0b31`; what remains is arithmetic, and the last section plans it.

1 Setup

Pipeline. Each benchmark runs through the runner `run-holes-rw2.sh` on one 8-core slot:

1. **cvc5** (branch `alethe-rewrite-granularity`, static `bin/cvc5-rw`) solves and prints an Alethe proof with the options
`--proof-granularity=rewrite --proof-alethe-res-pivots`
`--proof-elim-subtypes --print-arith-lit-token`.
A rewrite the post-processor cannot expand is printed as `(step t (c1 (= t t'))`
`:rule hole :args ("MACRO_REWRITE" "RW_REWRITE"))` (or `"MACRO_SR_PRED_INTRO"`),
premise-free. A step it neither expands nor tags stays an *untagged* hole the pipeline
does not attempt: `THEORY_INFERENCE_ARITH`, `MACRO_THEORY_REWRITE_RCONS_SIMPLE`, the
`TRUST_THEORY_REWRITE` of subtype elimination, `DIAMONDS`.
2. **Hoist + prune:** `carcara elaborate --pipeline hoist prune`. Repeated hole subproofs
are lifted to the top level (`ho*` steps) and duplicates dropped; this is where QF_LIA's 1.97 M
hole steps become 748 k distinct holes and QF_LRA's 1.24 M become 505 k (QF_UF loses 5%).
3. **Elaboration pass:** `carcara elaborate --pipeline hole --elaborate-hole-rewrites`
with eight isolated workers (one child process per hole), smallest goal first, the set-form list

encoding, the normalizer prepass, sort guards and shared-subterm abstraction at 16 nodes. A hole is *justified* when a certificate is extracted from the saturated e-graph, verified by an independent rule checker, and emitted as Alethe steps; otherwise it is *kept* with the worker’s reason, or *skipped* when the pass budget ran out first.

4. **Re-check** of the elaborated proof by `carcara check` with the RARE file, so that `rare_rewrite` steps check.

There is no separate checking pass. “Proved” is derived from the elaboration pass as justified + no certificate + checker rejected + killed after egglog: a hole whose worker died in the serialization or the search had its goal proved by egglog. The derivation is conservative in one direction only – check and reconstruction share the 60 s, so a goal a pure checking pass would have proved but that was killed *inside* egglog here counts as lost. The pass uses the RARE file `holes-rw2.rare`, 171 rules: cvc5’s `big.rare` of the time, including its 80 bit-vector and string rules, plus `ite-eq`, `distinct-false` and the eight Boolean absorption, flattening and duplicate rules over `:list` parameters. (The local file was trimmed to 91 rules after the upload, on 2026-09-24; the cluster’s copy, checked by checksum, is the 171-rule one, so every number here is with the larger file. The trimmed file roughly halves the fixed cost of a hole in local measurements.) Carcara is the static build of commit `3f982c71`.

Limits. Per-hole limits are hard kills of the worker; the pass budget is a deadline counted from the CLI’s start, after which the remaining holes are *skipped* and the proof is emitted with what was achieved.

stage	limit
cvc5 solve + proof	60 s (<code>--tlimit</code>), external kill at 75 s
hoist + prune	300 s (the proof is then checked as it stands)
elaboration pass	1,500 s per proof; 60 s and 6 GB per hole; external kill at 1,600 s
egglog	growth caps 120 M (arithmetic) / 20 M tuples, memory soft cap 5.4 GB
re-check of the elaborated proof	900 s
SLURM job	partition <code>octa</code> , 8 cpus, 60 GB, 3,100 s wall, two tasks per node

Benchmarks. The sets `rw_QF_UF` (4,316), `rw_QF_LIA` (2,541) and `rw_QF_LRA` (524): the benchmarks cvc5 proved within 60 s in the previous run of the pipeline (`rw1`, 2026-09-22/23), so no task is spent on a solver time-out and every proof is complete. The run took 152 task-hours on two tasks per node, about 1,217 core-hours reserved; the elaboration passes themselves sum to 139 hours.

2 Yield

Table 1 gives the yield per logic; Figure 1 the distribution of holes per proof after hoist and prune. `QF_UF` proofs are uniform (p50 253 holes, max 2k); `QF_LIA` has a fifth of its proofs with at most a dozen holes and a tail to 9k; `QF_LRA`’s tail reaches 15k, six proofs holding a fifth of the logic’s holes.

logic	proofs	hole steps	holes	p50	p90	max	every hole justified	re-check valid
QF_UF	4,316	1,288,303	1,221,095	253	469	2,062	1,907 (44.2%)	322
QF_LIA	2,541	1,973,773	747,899	15	643	8,763	2,263 (89.1%)	982
QF_LRA	524	1,238,485	504,702	64	2,662	15,544	326 (62.2%)	202

Table 1: Yield. “hole steps” are the `hole` steps in `cvc5`’s proof, “holes” the distinct goals left after hoist and prune, with the median (p50), 90th percentile and maximum per proof. “every hole justified” counts the proofs in which no attempted hole was kept or skipped, with the share of the logic’s proofs; “re-check valid” the proofs whose elaborated output passed the final check with no `hole` step at all.

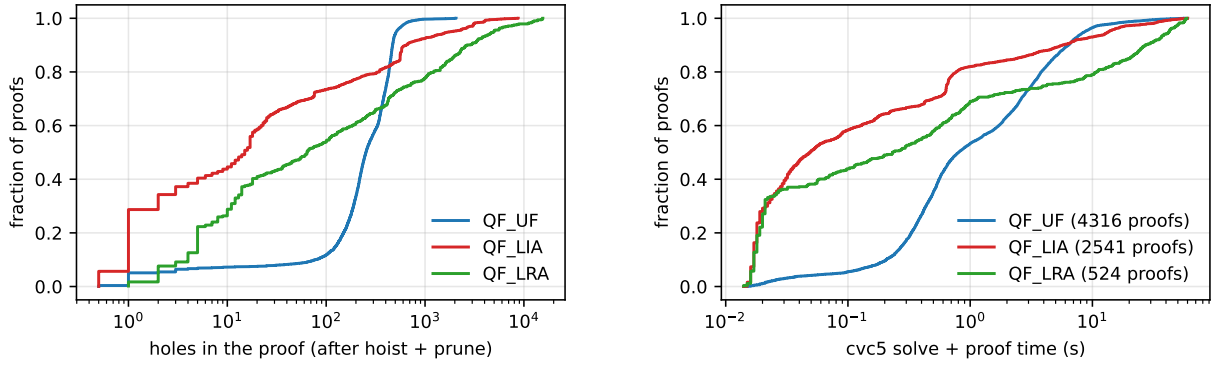


Figure 1: Left: holes per proof, as a cumulative distribution (log scale). Right: `cvc5`’s solve-and-print time; every proof is complete by construction of the sets.

outcome	QF_UF	QF_LIA	QF_LRA
re-check valid	322	982	202
re-check holey: every hole justified, untagged holes left	1,563	987	124
re-check holey: every hole justified, a trust step of the elaborator left	22	294	0
re-check holey: a hole kept or skipped	2,211	249	189
hoist + prune rejected the proof (resolution pivot)	198	2	6
hoist + prune timeout (300 s)	0	8	0
elaboration pass external limit (1,600 s)	0	16	3
task killed at the job memory limit (60 GB) before the pass ended	0	3	0

Table 2: Outcome of every benchmark. The 206 hoist rejections are `cvc5` resolution steps whose pivot is not in the clause (the QG-classification `iso` family in QF_UF, two Goel, four Heizmann, two LassoRanker and two SMPT proofs), a `cvc5` defect known from earlier runs. The 8 hoist time-outs continue with the unhoisted proof: four of them finish the pass with holes left, four hit its external limit. The external limit fires when the 1,500 s deadline expires inside a step the budget cannot interrupt; the passes it cut ran on proofs of 60 to 3,900 holes, and their work is lost with the process. Three QF_LIA tasks (`SpamAssassin-loop`, `prp-2-17`, `prp-3-18`) were killed at the job’s 60 GB during the pass and have no record, the same three that were lost the same way in the earlier runs.

3 Hole outcomes

Table 3 and Figure 2 give the outcome of every hole of the proofs that ran the pass. Three numbers matter:

- **Of the holes attempted**, elaboration justifies 99.5% (QF_UF, QF_LIA) and 99.3% (QF_LRA); counting the holes egglog proved but the reconstruction lost, the engine proves 99.96% / 99.7% / 99.5%. This is the engine’s rate on cvc5’s rewrite goals, and it no longer depends on the logic.
- **The budget no longer binds**. 543 holes were skipped in the whole run (Table 5), against tens of thousands in `rw1`; only QF_LRA still has proofs above 10k holes, and they finish. Figure 3 shows the outcome by proof size and Figure 4 the pass time against it: the pass scales with the hole count and stops at the budget in a handful of proofs.
- **The normalizer does a third of the work** (Table 4). Of the holes attempted, 30.0% are closed by the normalizer alone – both sides normalize to the same term under the four checker-certified procedures – and the largest part of the rest goes to egglog in normal form, where it is proved at 98–99% and bridged back to the stated goal. The retry of the stated goal after the normal form fails is worth keeping on QF_LIA (2.7k of 4.8k retries proved) and nearly worthless on QF_LRA (44 of 1.8k).

logic	holes	attempted	justified	kept	skipped	% of attempted	% of all	proved, lost after egglog	proved, % of all
QF_UF	1,171,831	1,171,798	1,166,150	5,648	33	99.5	99.5	5,144	99.95
QF_LIA	713,883	713,402	709,945	3,457	481	99.5	99.4	1,718	99.69
QF_LRA	433,860	433,831	430,962	2,869	29	99.3	99.3	513	99.45

Table 3: Hole outcomes of the pass, over the proofs that ran it. “attempted” = justified + kept; “skipped” = never tried because the budget ran out. “proved, lost after egglog” are the kept holes whose goal egglog had proved (no certificate, checker rejected, decode failure, or killed in the search or the serialization); the last column adds them to the justified ones.

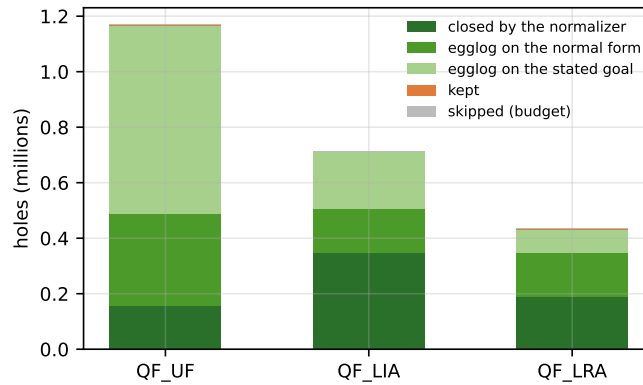


Figure 2: The same counts as a picture, with the justified holes split by who closed them: the normalizer alone, egglog on the normal form, or egglog on the stated goal.

logic	holes	closed by the normalizer	normal form to egglog	proved, bridged	retried as stated, proved	stated goal only	% justified
QF_UF	1,171,831	155,455 (13.3%)	339,484	333,594	387 of 5,890	676,714	99.5
QF_LIA	713,883	349,624 (49.0%)	160,387	155,549	2,671 of 4,838	202,101	99.4
QF_LRA	433,860	189,707 (43.7%)	160,464	158,616	44 of 1,848	82,595	99.3

Table 4: The normalizer’s share. “closed” are holes whose two sides normalize equal; “normal form to egglog” the holes handed to egglog as their normal form, of which “proved, bridged” came back with a certificate joined to the stated goal; “retried as stated” the ones whose normal form failed and whose stated goal was then tried; “stated goal only” the holes whose normal form was the goal itself.

logic	proofs run	hit 1,500 s	holes skipped there	hit the 1,600 s external limit	proofs >10k holes	% of holes
QF_UF	4,118	1	33	0	0	0.0
QF_LIA	2,516	11	481	20	0	0.0
QF_LRA	515	6	29	3	6	18.9

Table 5: Where the pass budget binds. “proofs run” are the proofs with a pass record.

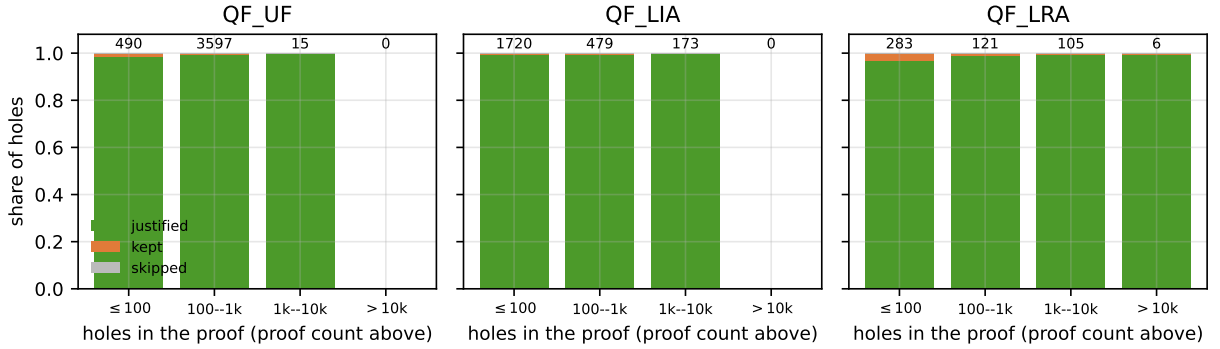


Figure 3: Outcome of the holes by proof size. The budget is visible only in QF_LIA’s and QF_LRA’s largest bucket, and there as a sliver.

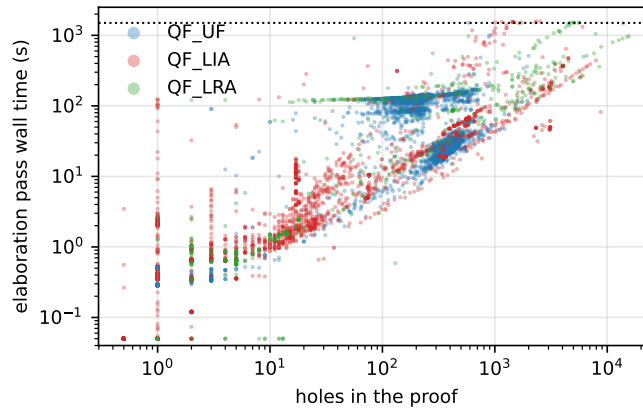


Figure 4: Pass wall time against proof size, one point per proof; the dotted line is the budget. The plateau at 60s in small proofs is a single hole running to its per-hole limit; the points above the budget are the passes the deadline could not interrupt.

logic	family	proofs	holes	% justified	% kept	% skipped
QF_UF	QG-classification	3,655	1,130,800	99.5	0.5	0.0
QF_UF	2018-Goel-hwbench	225	21,632	98.6	1.3	0.2
QF_UF	NEQ	41	12,970	100.0	0.0	0.0
QF_UF	PEQ	27	3,828	100.0	0.0	0.0
QF_UF	SEQ	36	2,260	100.0	0.0	0.0
QF_LIA	20220307-SMPT	1,518	177,392	99.9	0.1	0.0
QF_LIA	mathsat	94	139,129	100.0	0.0	0.0
QF_LIA	bofill-scheduling	245	116,673	100.0	0.0	0.0
QF_LIA	rings	84	69,618	99.6	0.4	0.0
QF_LIA	rings_preprocessed	90	65,035	100.0	0.0	0.0
QF_LRA	uart	18	113,026	99.5	0.5	0.0
QF_LRA	LassoRanker	65	95,495	99.9	0.1	0.0
QF_LRA	sc	25	79,557	98.0	2.0	0.0
QF_LRA	tta_startup	39	42,439	99.5	0.5	0.0
QF_LRA	clock_synchro	34	29,096	99.9	0.1	0.0

Table 6: The five families with the most holes per logic.

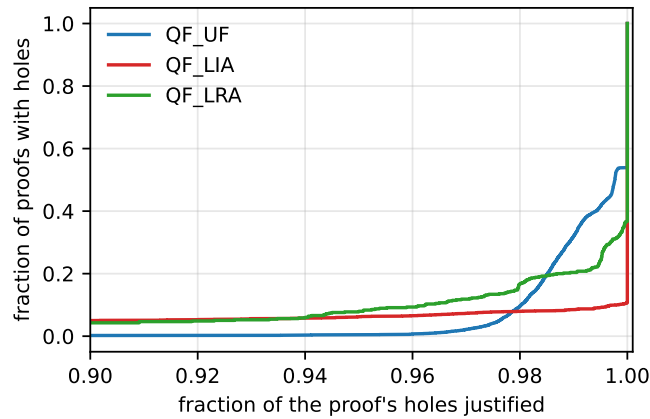


Figure 5: Per-proof share of holes justified (the axis starts at 0.9). In every logic more than 90% of the proofs have more than 99% of their holes justified; the tail is Dartagnan in QF_LIA and the four hole-heavy families in QF_LRA.

4 Per-hole cost

Table 7 and Figure 6 give the time per hole for the holes the pass justified, once over all of them and once over those that went through egglog (the normalizer’s closures cost nothing measurable). The QF_UF median of 0.50 s is a floor rather than a distribution: a worker is a child process that parses the rule file, builds the e-graph and saturates, and the coarse Boolean holes of QG-classification all cost the same. QF_LIA’s fat tail (115k holes over a second) is the arithmetic saturation on whole-assertion goals. Figure 7 and Table 8 split the time of the holes that went through egglog into its phases: saturation is 65% of it on QF_UF and 82% on the arithmetic logics, the certificate search 34% and 16–17%, the snapshot under 1%. The search’s share on QF_UF is the coarse Boolean holes again, and so are its kills: 2,875 QF_UF holes died in the search after egglog had proved the goal, against 469 inside egglog. Eight workers reach 3.5–5.4 cores of the 8: the parent’s parsing, hoisting and re-check are serial, and the smallest-first worklist drains into a few long holes at the end of a proof.

logic	holes	finished	p50	p90	p99	max	≥ 10 s	CPU hours
QF_UF	all justified	1,163,937	0.50	0.57	1.80	118.4	3,185	157.3
	through egglog	1,008,678	0.51	0.58	1.86	118.4	3,185	157.3
QF_LIA	all justified	686,232	0.31	1.26	4.65	115.3	3,020	129.7
	through egglog	348,691	0.86	1.42	7.33	115.3	3,020	129.7
QF_LRA	all justified	431,485	0.39	0.93	1.87	117.1	359	49.6
	through egglog	241,581	0.56	1.11	2.13	117.1	359	49.6

Table 7: Time per hole, holes the pass justified. The maxima above 60 s are holes whose worker was killed at the limit and retried on the stated goal after the normal form.

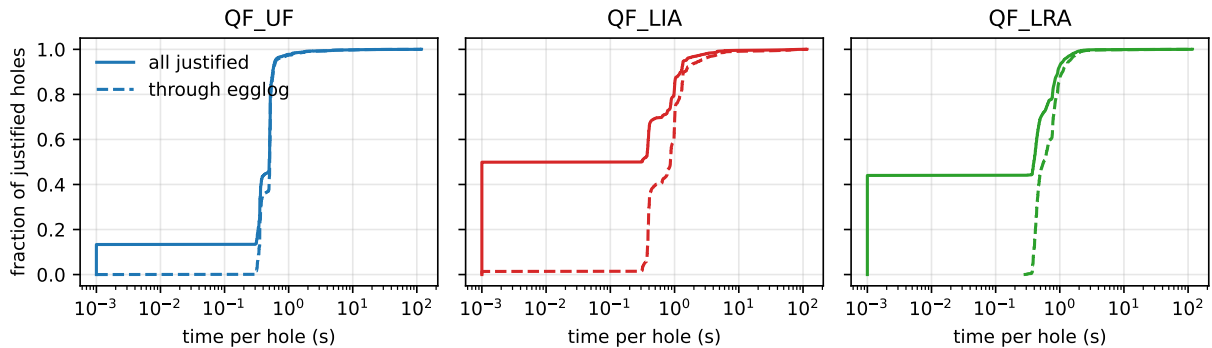


Figure 6: Per-hole time, all justified holes (solid) and those that went through egglog (dashed).

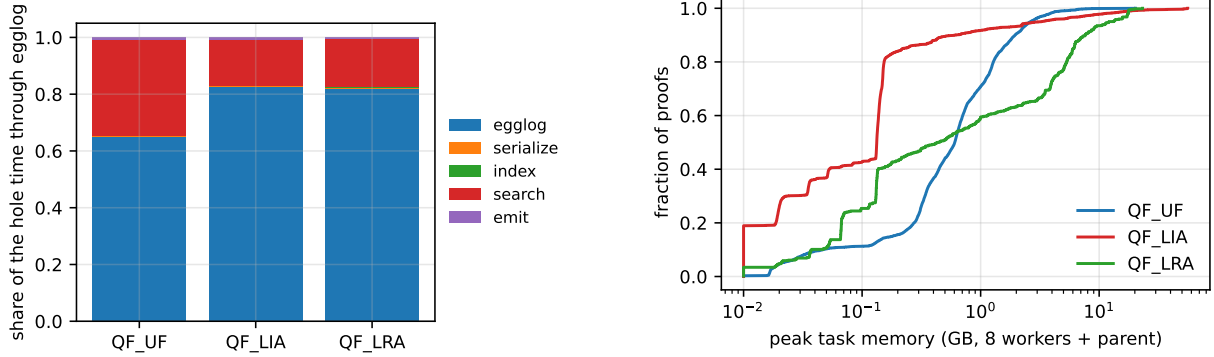


Figure 7: Left: share of the hole time per phase, holes that went through egglog. Right: peak memory of the task (parent plus eight workers, each limited to 6 GB).

logic	phase	holes	p50	p90	max	% of time	kills in phase
QF_UF	egglog	1,014,214	0.408	0.425	60.0	64.9	469
	serialize	1,014,214	0.001	0.002	2.2	0.4	2
	index	1,014,214	0.000	0.000	0.3	0.0	0
	search	1,014,214	0.086	0.134	59.2	33.7	2,875
	emit	1,014,214	0.001	0.006	8.2	1.0	8
QF_LIA	egglog	359,080	0.746	1.251	58.5	82.6	1,254
	serialize	359,080	0.002	0.005	2.1	0.3	0
	index	359,080	0.000	0.001	0.4	0.0	0
	search	359,080	0.094	0.120	56.3	16.3	253
	emit	359,080	0.001	0.003	17.1	0.8	4
QF_LRA	egglog	242,412	0.373	0.970	55.6	82.1	2,237
	serialize	242,412	0.002	0.004	4.2	0.4	0
	index	242,412	0.000	0.001	0.8	0.0	0
	search	242,412	0.098	0.141	51.1	17.1	26
	emit	242,412	0.001	0.003	2.8	0.4	0

Table 8: Elaboration phases per hole (seconds) and the phase a killed hole was in.

5 Why holes are kept

Table 9 and Figure 8 classify every kept hole by the worker’s reason, and Table 11 locates them by family. Three groups of about equal size:

- **Egglog time and memory kills** (35%). 3,814 holes hit the 60s limit inside saturation and 386 the 6 GB limit; 2,351 of them are in four QF_LRA families (`sc`, `uart`, `tta_startup`, `spider`) and 1,000 in QF_LIA’s Dartagnan. They are not all size: `sc`’s 1,509 kills are on goals of 59 and 114 nodes, a shape the rules blow up on (§7).
- **Reconstruction losses on QF_UF’s coarse Boolean holes.** 2,878 QF_UF holes were killed in the search after egglog had proved the goal, and 2,126 ended with no certificate; 137 more had their `and_simplify` step rejected by the checker (Table 10). These are the holes of QG-classification and Goel: Boolean simplification chains with many constant-valued arguments at once, which the search of 3f982c71 took one constant per congruence edge and one level per argument.
- **The honest residue and the checker’s rejections.** 261 goals egglog could not prove (`rings`, SMPT and `miplib`: a reflexive relation inside a disjunction that the normalizer closes but the pass had handed to egglog as stated), 177 Dartagnan `distinct_elim` steps the checker rejected, 46 certificates the elaborator could not decode, and 144 holes cut by the pass budget mid-flight.

reason	QF_UF	QF_LIA	QF_LRA	all
killed at the per-hole limit, inside egglog	457	1,158	2,199	3,814
killed at the per-hole limit, after egglog	2,878	248	26	3,152
no certificate	2,126	1,242	486	3,854
killed at the 6 GB limit	27	245	114	386
egglog could not prove	0	256	5	261
checker rejected	137	186	0	323
killed by the pass budget	8	98	38	144
certificate failed to decode	3	42	1	46
total	5,636	3,475	2,869	11,980

Table 9: Kept holes by reason. The runner files “egglog could not prove” (an egglog `Check failed` verdict) and the decode failures under `worker-error`; the split here is from the messages.

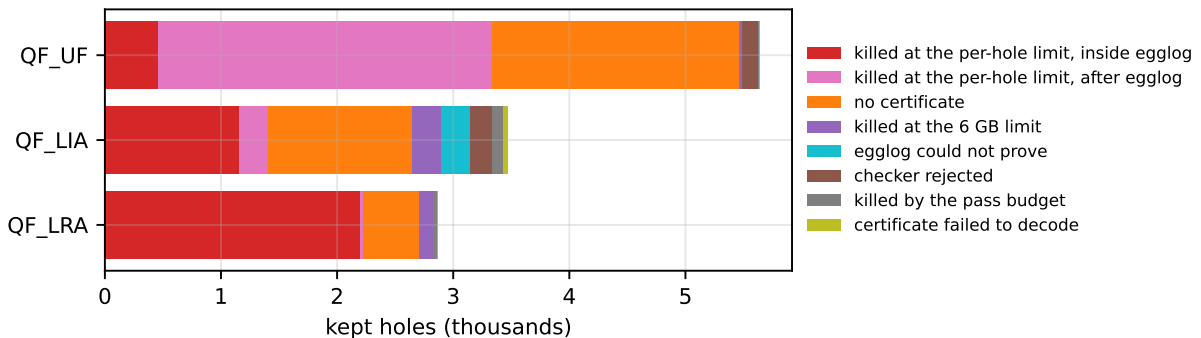


Figure 8: Kept holes by reason.

5.1 Reproduced and fixed after the run

The smallest proof of each class of Table 11 that is not an egglog time-out was replayed locally with the run’s options (notes §47.11, commits 36cc79fa and fedf0b31):

rule of the rejected step	QF_UF	QF_LIA	QF_LRA
<code>distinct_elim</code>	0	177	0
<code>and_simplify</code>	138	0	0
<code>resolution</code>	0	6	0
<code>or_simplify</code>	0	6	0

Table 10: Reconstructed steps the checker rejected, by rule. The `and_simplify` and `or_simplify` cases found the complementary pair through a nested connective while the rule reads direct arguments only; the `distinct_elim` cases are Dartagnan’s.

logic	family	proofs	every hole justified	kept	skipped	dominant classes
QF_UF	QG-classification	3,851	1,487	5,359	0	2,739 killed at the per-hole limit, after egglog, 2,103 no certificate
QF_UF	2018-Goel-hwbench	227	190	279	33	133 killed at the per-hole limit, after egglog, 88 killed at the per-hole limit, inside egglog
QF_UF	20190906-CLEARSY	11	5	8	0	6 killed at the per-hole limit, after egglog, 1 checker rejected
QF_UF	20170829-Rodin	20	18	2	0	2 certificate failed to decode
QF_LIA	20210219-Dartagnan	78	2	2,538	481	1,063 no certificate, 787 killed at the per-hole limit, inside egglog
QF_LIA	rings	84	49	288	0	190 egglog could not prove, 72 killed at the per-hole limit, inside egglog
QF_LIA	2019-ezsmc	13	0	212	0	212 killed at the per-hole limit, inside egglog
QF_LIA	calypto	13	0	167	0	98 no certificate, 49 killed at the per-hole limit, inside egglog
QF_LIA	20220307-SMPT	1,520	1,407	120	0	11 egglog could not prove, 9 killed at the per-hole limit, inside egglog
QF_LIA	fft	2	0	82	0	78 no certificate, 4 killed at the per-hole limit, inside egglog
QF_LRA	sc	25	0	1,597	29	1,509 killed at the per-hole limit, inside egglog, 50 no certificate
QF_LRA	uart	18	0	587	0	347 no certificate, 223 killed at the per-hole limit, inside egglog
QF_LRA	spider_benchmarks	42	6	283	0	166 killed at the per-hole limit, inside egglog, 68 killed at the 6 GB limit
QF_LRA	tta_startup	39	0	230	0	183 killed at the per-hole limit, inside egglog, 26 no certificate
QF_LRA	sal	96	61	79	0	52 killed at the per-hole limit, inside egglog, 20 killed at the 6 GB limit
QF_LRA	LassoRanker	67	46	57	0	41 killed at the per-hole limit, inside egglog, 11 no certificate

Table 11: The residue by family: the families with the most kept and skipped holes per logic.

- **Decode failures.** A congruence spine the search met from the far side arrives under `Symm` nodes, and the spine descent found no `cong` premises in it. It now reads `Symm` as a flip.
- **and_simplify / or_simplify rejections.** The flat form is now stated first by `aci_simp`, so the checker sees the complementary pair as direct arguments.
- **No certificate on Boolean chains with many constant arguments.** A candidate now replaces every constant-valued subterm at once; the ACI computation is the checker’s `aci_simp` in full; a one-step checker computation is tried before the congruence and the search; and no congruence is attempted between two different heads.
- **A goal egglog cannot prove as stated but the normalizer closes.** The rule that keeps a stated goal when its normal form is larger is confined to the checking pass; elaboration tries the normal form first whatever its size, the stated goal on a failure.

Every reproducer elaborates and re-checks. The Dartagnan proof `benchmark20_conjunctive-00` goes from 198 to 200 of 231 justified with no rejection or decode failure left; the smallest `rings` proof with unproved holes in the run justifies 349 of 349; `uart-5.base.cvc`, whose family had 347 no-certificate holes, keeps one of 299 (a memory kill inside egglog). Five larger samples keep their justified counts and run their passes in half the time (Table 12).

proof	3f982c71 (run binary)	fedf0b31
<code>clocksynchro_3clocks.main.invar.base</code>	195 of 196, 90 s	195 of 196, 65 s
<code>cut_lemma_01_008</code>	126 of 126, 17 s	126 of 126, 7 s
<code>MULTIPLIER_3.msats</code>	272 of 272, 16 s	272 of 272, 7 s
<code>tgc_io-safe-6</code>	140 of 140, 12 s	140 of 140, 5 s
<code>ring_2exp10_3vars_1ite.unsat</code>	349 of 349, 22 s	349 of 349, 11 s

Table 12: Five samples, holes justified and pass time, elaboration at 45 s per hole on four workers.

5.2 Limitations in the emitted proofs

Table 13 crosses “every hole justified” with the untagged holes and the re-check. 1,822 proofs are fully justified and carry no untagged hole, and 316 of them still re-check `holey`: a certificate whose `arith_poly_norm_rel` obligation the relation routing cannot express through `poly_simp_rel` is emitted as a `TRUST_THEORY_REWRITE` hole step (the `trusted` fallback of `rare_hole.rs`). That is 294 `QF_LIA` proofs, a quarter of the `QF_LIA` proofs the pipeline could have closed, and the cheapest `valid` gain left; the 22 `QF_UF` cases were not examined. The untagged holes themselves cap `valid` at 2,426 / 1,393 / 204 proofs and no Carcara change moves them.

logic	proofs with an untagged hole	every hole justified, untagged	every hole justified, no untagged	of those <code>valid</code>	of those <code>holey</code>	re-check timeouts
<code>QF_UF</code>	1,890	1,563	344	322	22	0
<code>QF_LIA</code>	1,148	987	1,276	982	294	21
<code>QF_LRA</code>	320	124	202	202	0	3

Table 13: Fully justified proofs against the untagged holes and the re-check.

6 The tooling at fedf0b31

Normalizer prepass (`elaborator/prenorm.rs`, driven from `elaborator/mod.rs`). Computes a normal form of every hole’s two sides with four checker-certified procedures: ACI normalization of `and` / `or` (`aci_simp`), polynomial normalization (`poly_simp`), relation normalization to a scaled difference with the constant on the right (`poly_simp_rel`), and `distinct` elimination. It is restricted to what core Alethe rules certify; no Boolean or equality simplification from the `*_simplify` family

lives here. Integer tightening is *not* implemented, although the specification in the notes (§25) lists it; §7 returns to this.

Engine (`rare/engine.rs`, `rare/computational/*`). RARE rules compiled to egglog with the goal’s terms as seeds; sort guard relations keep polymorphic rules on the right sorts; a conditional rule’s premise is demanded at the occurrence of its left-hand side rather than over all term pairs; `:list` parameters use the set-form encoding so flattening, absorption and duplicate rules fire in one step; polynomial normalization and evaluation are built-in relations; saturation runs in bounded steps against the growth caps, the memory soft cap and the deadline, and stops when egglog reports no update.

Reconstruction (`rare/reconstruction/`). A certificate search over the saturated e-graph from the goal’s left class to its right class: rule instances, congruence (one child or every constant-valued subterm replaced at once), computations (evaluation, ACI, polynomial, `distinct`), list rules on the set form, ACI-modulo matching; four rejustifications per obligation; every certificate verified by an independent rule checker before it is emitted.

Elaborator (`elaborator/rare_hole.rs`). Certificates become Alethe steps (`rare_rewrite`, `cong`, `trans`, `symm`, `refl`, `evaluate`, `aci_simp`, `and_simplify`, `or_simplify`, `poly_simp`, `poly_simp_rel`, `equiv_simplify`, `ite_simplify` and the rest of the checker’s computations), inserted at the hole with fresh ids and re-checked in the worker before the parent accepts them. Workers are isolated processes with a memory limit and a hard deadline; a kill is reported with its phase and the time spent before it.

Tests and files. 286 library tests, among them 13 end-to-end elaboration fixtures in `tests/rare/elaborate/`, each elaborated and re-checked valid. The static binary of `fedf0b31` is built and not yet uploaded.

7 What a rerun should change, and the three open directions

1. Rebuild the static binary from `fedf0b31` or later (§5.1), which removes the reconstruction losses of §5 on the replayed samples, and run the same three sets.
2. Count the trust steps the elaborator emits (§5.2) in the runner, and extend the relation routing so that they are not emitted.
3. Make the pass deadline interrupt the non-hole checking (23 external kills), and raise the re-check limit for the largest elaborated proofs (24 time-outs at 900s).

Dartagnan’s arithmetic Boolean chains (QF_LIA). 78 proofs, 2 fully justified, 3,019 holes lost; after the fixes, `benchmark20_conjunctive-00` still keeps 31 of 231 (22 no certificate, 4 memory, 2 time). The goals are cvc5’s rewrite of a whole assertion: an implication whose antecedent is a conjunction of bound pairs (`and (<= a b) (>= a b)`) and equalities (`= x 1`), rewritten into a flat conjunction of (`not (>= ...)`) / (`>= ... 1`) literals (the integer tightening of `<` and `>`), with `ite`-valued terms in the polynomials, 50–800 nodes; one 16-node case, (`= (ite c true true) true`) = `true`, has no certificate either, so at least one gap is a plain search gap. Plan: (1) micro-reproducers from the 22 goals, the `ite` case first (egglog proves it, so the search does not see the `ite-eq-branch` instance or discards the congruence into the `true` class); (2) a structural descent before the search – when both sides share the head, prove the children pairwise and join with `cong`, so that a whole-assertion goal becomes its 10–80 independent atom goals, each of which the search already proves; (3) a rejustification budget set from the goal’s atom count, if the debug trace of the 22 goals says four is the cap that bites; (4) for the egglog time and memory kills, the

per-ruleset run report on the smallest such goal, the polynomial rules on `ite` arguments being the first suspect; (5) a local batch of all 78 proofs (cvc5 solves each in under a second) as the exit criterion, 70 of 78 fully justified and no no-certificate class left, then the cluster run.

The hole-heavy QF LRA families. `sc`, `uart`, `tta_startup`, `spider`, `sal`, `LassoRanker`, `clock_synchro`: 2,852 holes lost, 2,351 of them egglog time or memory kills, and two different problems behind them. Small goals the rules blow up on: `sc`'s 1,509 kills are on goals of 59 and 114 nodes, and `sc-5.base.cvc` replayed locally (415 holes) keeps 14 – 7 goals of 59 nodes that fail *fast* (no rule path), 5 of 114 nodes that saturate for 60s, 2 without certificate – all `MACRO_SR_PRED_INTRO` disjunctions of transition conjunctions in which cvc5 flattened the `ands`, oriented the equalities, turned `<=` atoms into `(>= (+ ... ) 0.0)` mirrors and pushed `ites` into the polynomials; `uart`'s memory kill (t262, 179 nodes) is the same family one size down from its 333-node time-outs. And size: `tta_startup` (900–11,000 nodes), `LassoRanker` (3,000–8,700), `clock_synchro` (2,000–3,400) and `spider`'s 510-node memory kills, where saturation over the whole assertion does not finish. Plan: (1) the 59-node `sc` goal split into its disjuncts and conjuncts to find the pair egglog cannot join (candidates: the `<=` to `>=` mirror over a negated difference, which the normalizer does not orient and `comp_simplify` would certify; an equality against a normalized polynomial; `(= x (ite c a b))` lifting); (2) the 114-node goal profiled per ruleset with the `ite`-abstracted goal as control, and `ite` excluded from the polynomial relation's argument positions if that is the cause, as the normalizer already does; (3) the structural descent above for the big goals, each atom pair normalized first and only the differing pairs sent to egglog as their own small worker calls, plus a per-hole limit scaled by goal size; (4) the smallest proof of each of the seven families fully justified locally as the exit criterion, then the cluster run.

Integer infeasibility. The goal is `(= (+ (* 3 x1) (* 3 x2)) 1) = false` (QF_LIA check, `int_incompleteness1`; two holes in the run). Egglog has no rule path, since `arith-int-eq-conflict` needs the `(= (to_real t) c)` shape; the normalizer scales the relation by the reciprocal of the coefficients' gcd, producing `(= (+ x1 x2) 1/3)`, and stops, because the integer tightening its specification lists is not implemented. A hand-written proof of the closed form checks `valid` with the current checker: two `la_generic` steps, one per direction (`(c1 (not (= s 1)) (>= (+ x1 x2) 1))` with coefficients 1/3 and 1, and its mirror with `-1/3` and 1; the negation of the bound is strengthened by `la_generic`'s integer gcd rule), their resolution, and `equiv_simplify` with `equiv2` to state the equality. Plan: (1) implement the tightening in the normalizer's `relation_step` (a relation whose scaled bound is non-integral is decided or rounded, certified by the `la_generic` pair – a core rule, so within the normalizer's remit), which also gives it the `(> t 0)` to `(>= t 1)` step cvc5 applies throughout Dartagnan and SMPT; (2) alternatively, present an Int relation with a Real constant to the engine in the `to_real` shape so that the RARE tightening rules fire; (3) a fixture from `int_incompleteness1`'s hole; (4) both `check` benchmarks fully justified, and the count of `arith-geq-tighten` certificates in the Dartagnan and SMPT replays before and after.

8 Addendum: run `rw3`, and `cvc5`’s own expansion as the comparison arm

Two runs of 2026-09-25, read on 2026-09-26, on `rw2`’s benchmarks. `rw3` is the pipeline at commit `f0a15e53`: the fixes of §5.1, the trunk branch merged, the normalizer’s integer tightening and relation orientation, the structural descent of §7 at 32 nodes, the 91-rule file, and the re-check at 1,200s; everything else as in §1. `ds11` is new: the same `cvc5` binary at `--proof-granularity=ds1-rewrite` within 60s – every rewrite expanded by `cvc5`’s own RARE reconstruction into `rare_rewrite` steps – and the proof checked by `carcara check` with the 171-rule file (900s), no egglog, no elaboration. It answers what the egglog pipeline is competing with: the producer’s own expansion, checked by the same checker.

rw3 against rw2. Table 14 has the pass side by side. The orientation is what moved QF_LRA: the normalizer now closes 56% of its holes, the `sc` family’s saturation kills are gone, and the pass takes 13 hours instead of 21. The reconstruction fixes moved QF_UF: kept holes fall from 5,648 to 1,728 and the pass takes half the time. QF_LIA’s `valid` rises from 982 to 1,350 because the trust steps of §5.2 fell from 294 proofs to 11. The descent proved 7.6k / 20.2k / 33.9k holes and failed on 44.9k / 4.3k / 1.6k pairs; its cost shows in fifteen LassoRanker proofs fully justified in `rw2` and not in `rw3`, each lost to a few time kills after failed descents, and in QF_LIA’s skipped holes (643, from 481).

logic	run	holes	% justified	kept	skipped	% by the normalizer	every hole justified	of those, trust steps left	re-check <code>valid</code>	pass hours
QF_UF	<code>rw3</code>	1,171,831	99.85	1,728	0	13.27	2,907	661	675	48.8
	<code>rw2</code>	1,171,831	99.52	5,648	33	13.27	1,907	22	322	80.0
QF_LIA	<code>rw3</code>	705,120	99.49	2,963	643	51.51	2,350	11	1,350	38.2
	<code>rw2</code>	713,883	99.45	3,457	481	48.97	2,263	294	982	37.8
QF_LRA	<code>rw3</code>	433,860	99.75	1,074	0	56.28	325	0	203	12.8
	<code>rw2</code>	433,860	99.33	2,869	29	43.73	326	0	202	21.2

Table 14: `rw3` against `rw2`, holes of the proofs that ran the pass. “trust steps left” counts the fully justified proofs without an untagged hole that still re-check `holey` on a step the elaborator emitted.

Two defects of the elaborator surfaced in QF_UF and are fixed since (6824bb8a): 534 kept holes were a `cong` whose premises came out of argument order on a spine met from the far side, and 661 fully justified proofs re-checked `holey` on `TRUST_THEORY_REWRITE` steps emitted for engine-internal `gen-N` rewrites (the set-form identity removal, $(\text{and } A \text{ true}) = A$, and its kin) – the accounting gap of the first report, reopened by a search that now finds those edges. A `gen` step whose sides are equal modulo ACI is now the checker’s `aci_simp`; of twelve such proofs replayed, nine re-check `valid`, and the three left keep a conditional rule’s instance, which has no name in the program and stays trusted. The premises are sorted by the argument they justify. In QF_UF every trust step left is the elaborator’s (`cvc5` prints none there), so the honest QF_UF numbers of `rw3` are 2,907 proofs with every hole justified, 1,336 of them without an untagged hole, 675 `valid` – and most of the 661 lift with the fix.

ds11. Table 15. At this granularity `cvc5` loses 24 proofs to the 60s limit (10 QF_LIA, 14 QF_LRA: Dartagnan, LassoRanker) and prints *no* arithmetic preprocessing hole: the `THEORY_INFERENCE_ARITH` and `MACRO_THEORY_REWRITE_RCONS_SIMPLE` steps that cap the pipeline’s `valid` at rewrite granularity (§5.2) are expanded. What remains is QF_UF’s `THEORY_LEMMA`, in 1,789 proofs, which the pipeline does not attempt either. The check errors are the 205 resolution-pivot proofs (198 / 2 / 5, the `cvc5` defect of Table 2) and 114 proofs citing `bool-implies-or-distrib`, which neither rule file had; the rule has since been added to both files, and all 114 proofs, regenerated locally, check `valid` with it. The four QF_LIA check time-outs are Dartagnan proofs of 5–12 MB at 900s.

logic	benchmarks	complete	valid	holey	error	time-out	holes left (proofs)	rule missing	cvc5 s, p50 / p90	check s, p50 / p90	check hours
QF_UF	4,316	4,316	2,230	1,789	297	0	4,149 in 1,789	99	0.98 / 6.9	0.17 / 1.0	0.64
QF_LIA	2,541	2,531	2,510	0	17	4	0 in 0	15	0.09 / 6.7	0.02 / 0.7	2.35
QF_LRA	524	510	505	0	5	0	0 in 0	0	0.40 / 29.6	0.08 / 2.5	0.11

Table 15: cvc5 at `dsl-rewrite` granularity, checked by Carcara. “holes left” are the `hole` steps in the complete proofs, all `THEORY_LEMMA`; “rule missing” the proofs citing a rule the 171-rule file lacked.

Head to head. Table 16 and Figures 9–10 put the two on the same benchmarks. A proof counts when nothing trusted is left: `valid` on both sides. Three arms of the pipeline are drawn: with elaboration, a proof that re-checks `valid`; with elaboration, a proof with every hole justified (untagged holes allowed, the engine’s own ceiling); and checking only, an *estimate* from the same run – the elaboration pass with the reconstruction phases of its holes (serialize, index, search, emit) taken out, summed over the holes and divided by the eight workers, and no re-check – since `rw3` folded the checking pass into the elaboration pass. The estimate keeps the pass’s parsing, the checking of the non-hole steps and every egglog phase, which is what a checking-only pass does; a measured checking-only run is the way to firm it up.

logic	benchmarks	dsl1 complete	dsl1 valid	rw3 every hole justified	rw3 valid	dsl1 only	rw3 only	dsl1 s, p50 / p90	rw3 s, p50 / p90	rw3 check-only s (est.), p50 / p90	hours, dsl1 / rw3 / est.
QF_UF	4,316	4,316	2,230	2,907	675	1,555	0	1.2 / 7.9	29.8 / 99.0	24.4 / 90.4	4.0 / 53.8 / 46.9
QF_LIA	2,541	2,531	2,510	2,350	1,350	1,160	0	0.1 / 7.6	2.3 / 95.3	2.2 / 90.4	4.1 / 42.7 / 38.7
QF_LRA	524	510	505	325	203	302	0	0.5 / 33.2	31.2 / 241.5	27.7 / 226.7	1.1 / 14.0 / 13.3

Table 16: The same benchmarks: proofs with no hole left, and the time from benchmark to checked proof (cvc5 plus check for `ds11`; cvc5, hoist, elaboration pass and re-check for `rw3`; the checking-only estimate as described).

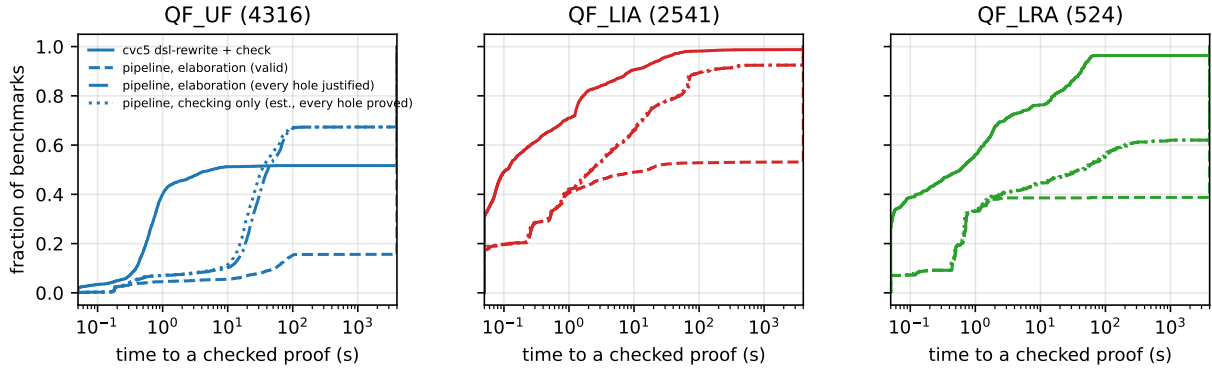


Figure 9: Time to a checked proof, as a cumulative distribution over the benchmarks of each set; a benchmark an arm does not finish is drawn at the right edge, so a curve’s height at the edge is the arm’s coverage. Solid: cvc5 at `dsl-rewrite` plus the check. Dashed: the pipeline with elaboration, proofs re-checked `valid`. Dash-dotted: the pipeline with elaboration, every hole justified (untagged holes allowed). Dotted: the pipeline checking only, the estimate, every hole proved.

cvc5’s expansion closes 5,245 of the 7,381 proofs with nothing trusted left against the pipeline’s 2,228, and every proof the pipeline closes it closes too (3,017 are its alone); it does so in 9.2 hours of cvc5-plus-check against the pipeline’s 110.6 with elaboration and about 98.9 checking only, a median of a second against tens of seconds. The gap is not the engine: `rw3` justifies 99.5–99.85% of the holes it is given and 5,582 proofs have every hole justified. It is the untagged holes cvc5 prints at rewrite granularity and expands at `dsl-rewrite`, plus the fixed cost of a saturation per hole. What the pipeline has that the expansion does not is independence from the producer’s reconstruction: a hole is closed from the rule set and the checker’s own normal forms, which is the case the first sections make; on the producer’s own proofs, the producer’s expansion is the cheaper way to a `valid` verdict.

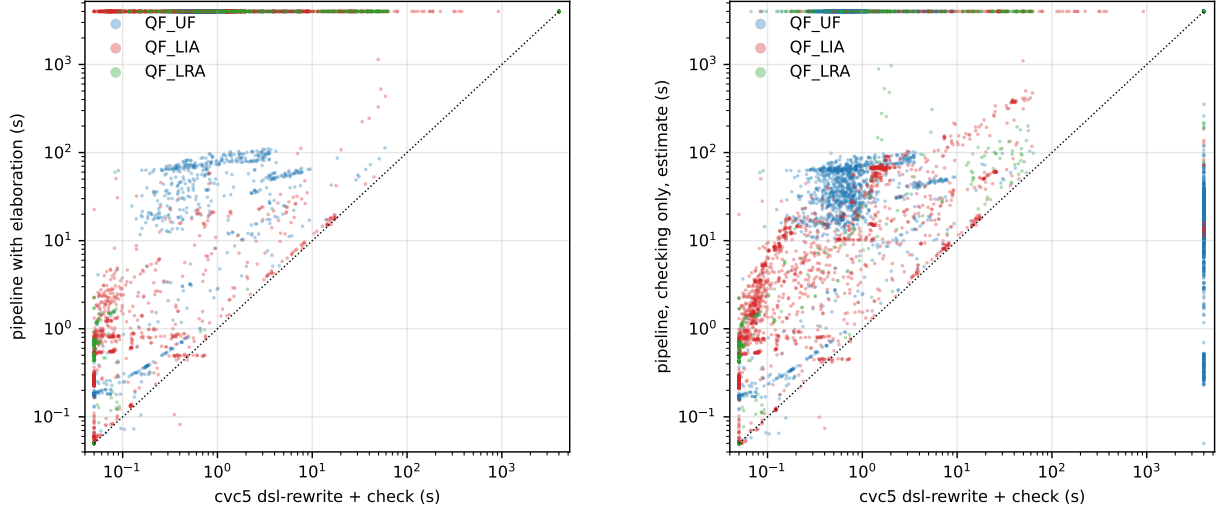


Figure 10: Per proof: cvc5 at `dsl-rewrite` plus the check (x) against the pipeline with elaboration (left, y : a proof re-checked `valid`) and against the checking-only estimate (right, y : every hole proved). A proof an arm does not finish sits on the edge; the dotted line is $y = x$.

9 Addendum: runs `rw4` and `rw5`

Two more runs on `rw2`'s benchmarks with `rw3`'s settings and a 92-rule file: `rw3`'s 91 rules and `bool-implies-or-distrib`, the rule `dsl1` missed. `rw4` (2026-09-27, commit `e6bc8ae4`) attempts cvc5's arithmetic preprocessing holes too (`THEORY_INFERENCE_ARITH`, `ARITH_STATIC_LEARN`, `MACRO_THEORY_REWRITE_RCONS_SIMPLE`, `SUBTYPE_ELIMINATION`: the untagged holes of §5.2) and carries the fixes of a local corpus built from `rw3`'s residue: relation keys over all terms and a relation mirror, a one-to-many and an atom alignment in the structural descent, every worker goal in a fresh process on a share of its hole's budget, and lambda applications tallied out of scope. `rw5` (2026-09-27/28, commit `7ec65b7d`) fixes what `rw4` left:

- the holes the search found no certificate for: absorbed `and/or` literals, arithmetic modulo the classes of its atoms, and ACI steps emitted in the shape the checker's `aci_simp` takes;
- `rw4`'s 126 checker rejections: an empty `cong`, premises placed by one side only, singleton connectives in rule steps, and a concluding step read off the wrong line;
- the search on a DAG: hash-consed terms and memoized walks, and every text a term is printed into (certificates, hole inputs, the egglog program) naming its shared subterms, where a goal of 188 nodes as a DAG and 247,127 as a tree had killed the search;
- an `and` equal to `false`, or an `or` equal to `true`, proved from one element of that constant's class, where the search lost the two-step path inside the saturated class;
- egglog's join order: the vendored egglog 0.4 binds a rule's pattern variables in order of the atoms they occur in, and the encoding repeats a leaf's atom at every use of the leaf, so leaves were bound before any atom related them, a cross product; a variable connected to a bound one now comes first (`third-party/egglog-0.4.0/CARCARA-PATCHES.md`);
- the set form's membership relation, through which the list rules match a set's own elements instead of every term of the e-graph, and no splice of a nested `and` through the class of a constant;
- a relation step between ground sides emitted as two `evaluate` steps instead of a trusted one.

Before it was submitted, **rw5** justified every hole of twelve replayed proofs (eleven **valid**, one **holey** on its **DIAMONDS** lemma) and every checked hole of the 49-slice corpus but one, which egglog saturates without proving.

The runs side by side. Table 17. **rw4**’s gain is breadth: the arithmetic tags add 196k **QF_LIA** and 87k **QF_LRA** holes to the pass, the corpus fixes take **QF_LIA**’s **no-certificate** and **unproved** holes from 985 and 789 to 8 and 68 (Table 18), and **valid** proofs rise from 2,228 to 4,194. **rw5**’s gain is cost: the pass takes 49.1 hours instead of 96.9 (**QF_UF** 48.5 to 15.9), the largest **QF_UF** task needs 8 GB instead of 24, the holes lost to time fall from 2,840 to 1,210, and **valid** proofs rise to 4,679 (1,828 / 2,404 / 447), with 6,586 proofs having every hole justified. Since **rw4** no fully justified proof re-checks **holey** on a step of the elaborator’s own.

logic	run	holes	% justified	kept	skipped	% by the normalizer	every hole justified	of those, trust steps left	re-check valid	pass hours
QF_UF	rw5	1,171,831	99.95	545	0	13.27	3,718	0	1,828	15.9
	rw4	1,171,831	99.88	1,452	0	13.27	3,283	0	1,437	48.5
	rw3	1,171,831	99.85	1,728	0	13.27	2,907	661	675	48.8
QF_LIA	rw5	899,511	99.89	978	13	61.05	2,421	0	2,404	27.6
	rw4	901,413	99.75	1,633	638	61.96	2,380	0	2,370	36.7
	rw3	705,120	99.49	2,963	643	51.51	2,350	11	1,350	38.2
QF_LRA	rw5	522,311	99.98	130	0	63.51	447	0	447	5.6
	rw4	520,546	99.95	284	0	63.46	387	0	387	11.7
	rw3	433,860	99.75	1,074	0	56.28	325	0	203	12.8

Table 17: **rw5** and **rw4** against **rw3**, holes of the proofs that ran the pass, as Table 14. **QF_LIA** and **QF_LRA** have more holes from **rw4** on: the arithmetic preprocessing holes are attempted.

logic	run	hole-time	memory	unproved	no-certificate	checker-rejected	out-of-scope	pass-budget	other
QF_UF	rw5	441	48	29	27	0	0	0	0
	rw4	1,355	18	0	12	67	0	0	0
	rw3	1,110	37	0	47	534	0	0	0
QF_LIA	rw5	643	54	91	2	3	212	23	0
	rw4	1,249	3	68	8	27	212	88	0
	rw3	952	255	789	985	8	0	80	0
QF_LRA	rw5	126	4	0	0	0	0	0	0
	rw4	236	2	0	14	32	0	0	0
	rw3	580	10	0	484	0	0	0	0

Table 18: Kept holes by the class of their reason. **out-of-scope** are lambda applications left unreduced; **pass-budget** holes the pass did not reach.

Hole by hole. Table 19 follows every hole of the proofs both runs finished through its outcome in each (hole ids agree: the **cvc5** proofs and the **hoist** are the same). 2,078 holes **rw4** kept are justified in **rw5**, most of them time kills (1,833) and **rw4**’s checker rejections (126); 187 change class and stay kept, mostly time kills that now end in egglog’s verdict or its memory limit, egglog getting further within the 60s; and 225 holes **rw4** justified are kept in **rw5**. Of these, 200 are in **Dartagnan**: 161 time kills in 8 proofs, 109 of them in **in-de20** (fully justified in neither run) and 38 in **seq-3**, 21 pass-budget skips and 18 memory kills (**sendmail-close-angle**, **seq-3**, **theatreSquare**); 21 are time kills in **clock_synchro**, **SMPT** and **LassoRanker**, one or two per proof. Of the time kills whose **rw4** time is logged, 115 took 40s or more there and 15 took 20–40s: holes at the 60s limit. The last four, in **rings**, are a completeness regression: **rw4** proved them in under 5s, and **rw5**’s egglog saturates without the goal; they, the memory kills and **theatreSquare**’s three new checker rejections are being looked at. The 18 proofs **rw5** loses to **rw4** are 10 **clock_synchro** and 5 **SMPT** proofs, each on one time kill, the two **rings** proofs, and one **CIRC** proof whose **cvc5** run aborted at its 60s limit.

	rw4	rw5	holes	QF_UF	QF_LIA	QF_LRA
now justified	hole-time	justified	1,833	821	891	121
	checker-rejected	justified	126	67	27	32
	pass-budget	justified	57	0	57	0
	no-certificate	justified	34	12	8	14
	unproved	justified	19	0	19	0
	memory	justified	9	7	0	2
kept in both, the class changed	hole-time	memory	70	37	29	4
	hole-time	unproved	57	29	28	0
	hole-time	no-certificate	29	27	2	0
	pass-budget	hole-time	28	0	28	0
	hole-time	checker-rejected	3	0	3	0
justified in rw4 , kept in rw5	justified	hole-time	182	0	167	15
	justified	pass-budget	21	0	21	0
	justified	memory	18	0	18	0
	justified	unproved	4	0	4	0

Table 19: Holes whose outcome differs between **rw4** and **rw5**, on the proofs whose elaboration pass finished in both; transitions of fewer than three holes are left out.

Against cvc5’s expansion. Table 20 and Figure 11 repeat Table 16 and Figure 9 for **rw5**. **ds11** closes 5,245 proofs with nothing trusted left against **rw5**’s 4,679 (**rw3**: 2,228); 576 are **ds11**’s alone and 10 **rw5**’s. The untagged holes no longer explain the gap: 567 of the 576 have a hole **rw5** kept or skipped, and the proofs **rw5** fully justifies and cannot call **valid** (1,890 in QF_UF, 17 in QF_LIA) carry **THEORY_LEMMA** steps (1,789, **holey** in **ds11** as well), **DIAMONDS** steps (114) or, in three proofs, another untagged step; one QF_LIA re-check ran out of time. The gap in cost narrows and stays: 60.7 hours from benchmark to checked proof against **ds11**’s 9.2 (**rw3**: 110.6), about 54.7 checking only, and medians of 2–12s against 0.1–1.2s.

logic	benchmarks	ds11 complete	ds11 valid	rw5 every hole justified	rw5 valid	ds11 only	rw5 only	ds11 s, p50 / p90	rw5 s, p50 / p90	rw5 check-only s (est.), p50 / p90	hours, ds11 / rw5 / est.
QF_UF	4,316	4,316	2,230	3,718	1,828	402	0	1.2 / 7.9	12.4 / 46.9	10.8 / 44.3	4.0 / 20.9 / 19.0
QF_LIA	2,541	2,531	2,510	2,421	2,404	108	2	0.1 / 7.6	1.8 / 67.9	1.7 / 64.8	4.1 / 32.9 / 29.4
QF_LRA	524	510	505	447	447	66	8	0.5 / 33.2	6.4 / 151.6	5.8 / 139.4	1.1 / 6.9 / 6.3

Table 20: **rw5** against **ds11** on the same benchmarks, as Table 16.

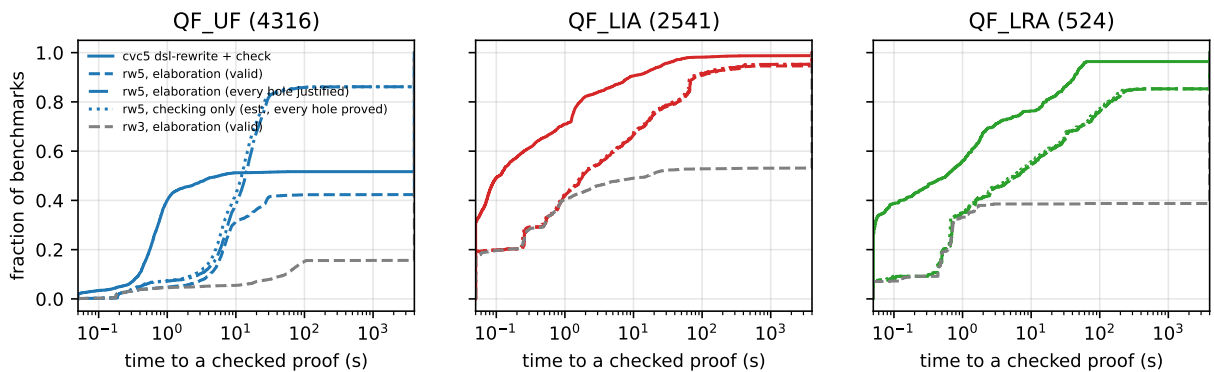


Figure 11: Time to a checked proof, as Figure 9 for **rw5**; grey dashed: **rw3** with elaboration, proofs re-checked **valid**.

A Files

- Results: `~/exp/results/egglog-holes/rw2/results.json.gz` (JSONL; one record per benchmark with the runner’s full output); **rw3**/ and **ds11**/ likewise for §8, **rw4**/ and **rw5**/ for

§9.

- Analysis: `report-rw2/make-rw2.py` in the worktree, which produced every table and figure of §1–§7, `make-rw3.py` those of §8, and `make-rw5.py` those of §9; `~/exp/egglog-holes/analysis-rw2/` (and `analysis-rw4/`, `analysis-rw5/`) hold the readers behind the notes.
- Runner and submission: `~/exp/egglog-holes/run-holes-rw2.sh`, `submit-egglog-rw2.sh`, `gen-rw-sets.py`; `run-holes-rw4.sh` and `run-holes-rw5.sh` with their submit scripts for §9.
- Notes: `EGGLOG-ELABORATION-NOTES.md` §47.9–47.13 in the Carcara worktree, §47.14–47.30 for the addenda; the replays of §5.1 and §7 are in the session scratchpad (`rw1/werr`, `rw1/lra`, `gcd`).